The empirical recurrence for OEIS A251039, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A251039 counts arrays over $\{0, 1, 2, 3\}$ in which a statistic taken over short runs of cells produces a derived array that is monotone in a named direction. The entry carries an empirical recurrence of order 32 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: every comparison the condition makes lies inside a bounded window of consecutive rows, so the count is a walk count on $S = 64$ states, and the conjectured recurrence is then settled by a finite exact computation.

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1 The conjecture

OEIS A251039 is "Number of (2+1)X(n+1) 0..3 arrays with nondecreasing min(x(i,j),x(i,j-1)) in the i direction and nondecreasing absolute value of x(i,j)-x(i-1,j) in the j direction.". It has offset 1 and begins

$$636, 6089, 45560, 308423, 1907070, 11102613, 61631550, 329986925, \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 17*a(n-1) -94*a(n-2) +47*a(n-3) +1342*a(n-4) -3529*a(n-5) -6251*a(n-6) +32794*a(n-7) +3826*a(n-8) -154583*a(n-9) +78628*a(n-10) +449757*a(n-11) -384214*a(n-12) -871126*a(n-13) +949876*a(n-14) +1162304*a(n-15) -1489818*a(n-16) -1081492*a(n-17) +1584172*a(n-18) +698807*a(n-19) -1168177*a(n-20) -306648*a(n-21) +597764*a(n-22) +86772*a(n-23) -208654*a(n-24) -13947*a(n-25) +47793*a(n-26) +736*a(n-27) -6664*a(n-28) +104*a(n-29) +484*a(n-30) -13*a(n-31) -12*a(n-32)

As of the "Last modified" line on the live entry (Jul 23 2025 13:02:01, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 3 cells across and entries $x(i, j) \in \{0, 1, 2, 3\}$.

Two derived arrays are formed. The first reads the horizontal statistic across each neighbouring pair of a row,

$$b(i, j) \;=\; \big[\, |a - b| \,\big]_{a=x(i,j),\, b=x(i,j-1)} \qquad (j \geq 1),$$

and must not decrease DOWN a column: $b(i, j) \leq b(i+1, j)$. The second reads the vertical statistic down each neighbouring pair of a column,

$$c(i, j) \;=\; \big[\min(a, b)\big]_{a=x(i,j),\, b=x(i-1,j)} \qquad (i \geq 1),$$

and must not decrease ALONG a row: $c(i,j) \le c(i,j{+}1)$. In words, the horizontal statistic is the absolute difference of a cell and its left neighbour, and the vertical one is the smaller of a cell and the cell above it. The entry writes the array with n as the number of COLUMNS. The walk is run down the transpose, which exchanges the roles of the two directions once and for all; the arguments of each statistic keep their order under that exchange, the later cell staying the first argument.

Nothing in the condition is invariant under renaming the letters — each statistic either does arithmetic on them or compares them by size — so no reduction to equality patterns is available, and the walk is run over the alphabet the entry names.

3 A bounded window

Lemma 1. *Both conditions are conditions on TWO consecutive array rows and nothing more.*

Proof. $b(i,j) \le b(i{+}1,j)$ names four cells, all in rows i and $i+1$. $c(i,j) \le c(i,j{+}1)$ names four cells, all in rows $i-1$ and i. No comparison reaches further. ☐

So a single row is a state. Let M be the matrix on the $4^3 = 64$ rows with $M[r,s] = 1$ when the pair (r,s) passes both tests and 0 otherwise. An array of R rows is a walk of $R-1$ steps and every row may begin or end one, so with $\iota = \tau = \mathbf{1}$,

$$a(n) \;=\; \iota^\top M^{\,j} \tau, \qquad S = 64.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^{32} - \sum_i c_i t^{32-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+32} - \sum_i c_i M^{j+32-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. ☐

Theorem 3.

$$a(n) \;=\; 17\,a(n{-}1){-}94\,a(n{-}2){+}47\,a(n{-}3){+}1342\,a(n{-}4){-}3529\,a(n{-}5){-}6251\,a(n{-}6){+}32794\,a(n{-}7){+}3826\,a(n$$

for every $n > 31$.

Proof. The vector $w = q(M)\tau$ was formed by 32 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 31 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. ☐

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: fill the cells in row major order and test each comparison the moment both of its runs are complete, in the orientation the entry's own name writes. No transfer matrix and no transposition are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A251039.

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[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.